SLIDE 1

Problem & why it matters

TEAM DengueSUX **PROBLEM STATEMENT** A2 — DengueRadar: Real-time outbreak forecasting

THE PROBLEM IN ONE SENTENCE

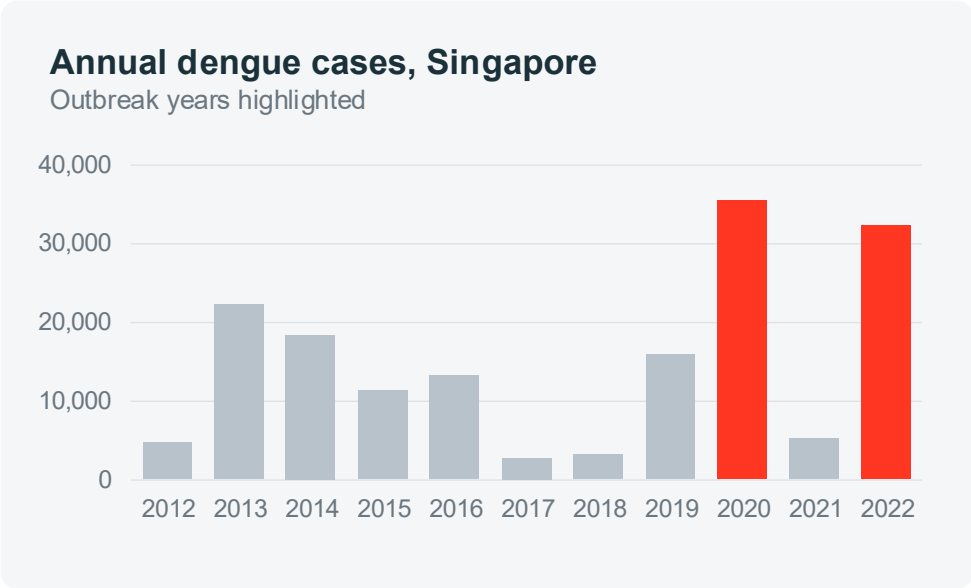
Singapore's public view is strongest after cases are reported. DengueSUX makes earlier weather and cluster context visible while stating exactly when a current national forecast is unavailable.

WHO IS AFFECTED, AND HOW BADLY

164,447 dengue cases in Singapore from 2012–2022, including a record **35,315 in 2020;** 1,792 in a single week at the peak. Repeat infections carry a higher risk of severe dengue.

WHY IT MATTERS NOW

Outbreaks swing more than 10× between years (2,776 cases in 2017 vs 35,315 in 2020). Weather can provide leading context, but strain changes and reporting delays matter too. We therefore show a two-week backtest, a freshness-gated live mode, and dated local signals.



Source: MOH Weekly Infectious Disease Bulletin, data.gov.sg (Dengue Fever + DHF)

"I've had dengue three times. The third time, I almost died." — why we're building this

Solo Project **Apollo Joshua** · Nanyang Technological University · Computer Engineering · Year 2 · POLO0002@e.ntu.edu.sg

SLIDE 2

Solution & data

OUR SOLUTION IN ONE SENTENCE

DengueSUX turns open data into a weekly plan: **which areas to check first, which blocks to walk, and what to tell residents**, backed by a 2-week national forecast tested on the 2022 outbreak.

WHO USES IT, AND WHAT DECISION IT CHANGES

Town council officers and grassroots volunteers decide **where to inspect this week and in what order**, starting from 8 ranked stops instead of a whole estate. Residents get a notice in their own language.

WHAT MAKES IT DIFFERENT FROM A GENERIC DASHBOARD

Street-level inspection plan

Real cluster blocks → 8 places to check on a 1.2 km OneMap walking route.

Tested on a real outbreak

About 2× more accurate than a simple average on 2022; turned High 11 weeks before the peak.

AI Analysis

Top drivers and a dated source on every figure. Places to check for fumigation and maintenance suggestions.

Speaks residents' languages

Notices in English, 中文, Melayu and தமிழ், each passing 5 automatic fact checks.

DATASETS (ALL OPEN AND PUBLIC)

1. **Weekly Infectious Disease Bulletin**
   MOH · 574 weeks of dengue cases, 2012–2022
2. **Rainfall, temperature & humidity archives**
   NEA · 125M station readings, 2017–2022
3. **Real-time weather APIs (V2)**
   NEA · live station readings
4. **Dengue Clusters (GeoJSON)**
   NEA · live; archived daily since 24 Sep 2026
5. **Master Plan 2025 Planning Areas**
   URA · 55 areas
6. **HDB Property Information**
   HDB · matches cluster block addresses
7. **Parks · Hawker Centres**
   NParks, NEA · candidate places to check
8. **OneMap Search & Routing**
   SLA · block locations and walking routes

Honest limits: public case data ends in 2022, so the forecast replays 2022 until recent weekly counts are approved. Area ranks are priorities, not local case predictions.

SLIDE 3

Databricks architecture & impact

SOURCE
data.gov.sg + OneMap
Bulletin, weather, clusters, URA, HDB, parks, hawkers

→

INGEST
Bronze Delta
Checksummed raw files; daily cluster snapshot job

→

TRANSFORM
PySpark → Delta
Sun–Sat epi-weeks, 55-area panel, cluster → block → place joins

→

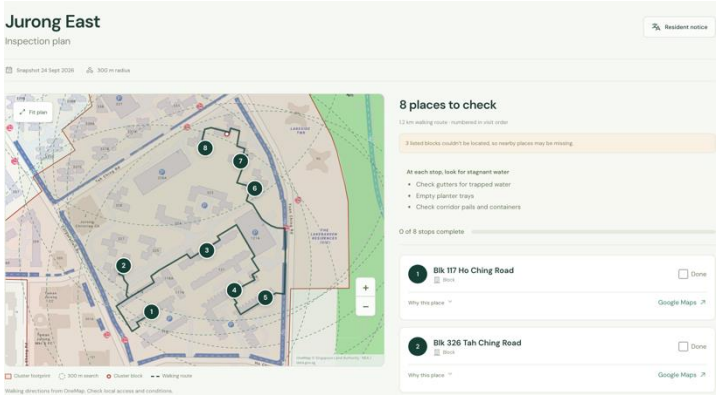
MODEL
MLflow
3 models, 39 folds (2019–21); gradient boosting won; SHAP drivers

→

OUTPUT
App + Genie
Map, inspection routes, 4-language notices via Foundation Model API

UNITY CATALOG governs every table: source receipts and checksums, lineage from raw file to forecast and route, access controls

WHAT WE WILL DEMO



Already working: 8 places to check on a 1.2 km walking route around the largest cluster (real NEA, HDB, OneMap data).

MEASURABLE IMPACT

- Forecast off by ~83 people a week vs ~172 for a simple average: 52% lower error on the 2022 outbreak
- Turned High 11 weeks before the 2022 peak, 1 week earlier than the simple average
- Teams start with 8 ranked stops instead of a whole estate
- 220 ready-to-review resident notices: 55 areas × 4 languages

WHAT WE CAN FINISH IN TWO WEEKS

- Deploy the 6 notebooks and daily job on Databricks Free Edition (serverless), with Unity Catalog tables and MLflow
- Replace the scripted prototype assistant with a Genie space
- Run the forecast live once recent weekly case counts are approved
- Switch on cluster-growth ranking after 7+ days of snapshots